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102
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Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Engineering Physics | **Subject Code : BSC 102**

Time Allowed : 3 Hours | **Full Marks : 70**

Pass Marks : 21

Answer in your own words.

Answer any five *questions in which* Question No. 1 *is compulsory.*

The figures in the margin indicate full marks.

All questions carry equal marks.

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1. Answer the following multiple-choice questions: 2×7

(i) Which instrument is used to measure very small thickness?

(a) Vernier calipers (b) Screw gauge

(c) Micrometer (d) Compass

(ii) Which of the following is a vector quantity?

(a) Speed (b) Distance

(c) Acceleration (d) Work

(iii) Dimensional formula of force is

(a) $[MLT^{-2}]$ (b) $[ML^2T^{-2}]$

(c) $[MT^{-2}]$ (d) $[LT^2]$

(iv) Which law states that stress is proportional to strain within elastic limit?

(a) Pascal's law (b) Hooke's law

(c) Boyle's law (d) Newton's law

(v) Which mode of heat transfer occurs in solids?

(a) Conduction (b) Convection

(c) Radiation (d) All of these

(vi) The acceleration due to gravity decreases with

(a) Altitude (b) Depth

(c) Latitude (d) All of these

(vii) The frequency of a wave is 500 Hz and wavelength is 0.68 m. Find its velocity.

(a) 340 m/s (b) 300 m/s

(c) 250 m/s (d) 400 m/s

2. (a) Define fundamental and derived quantities with examples.

   (b) Convert 5 m/s velocity into cm/s. 7+7

3. (a) Explain Newton's third law of motion with practical examples.

   (b) A body moving with initial velocity 10m/s accelerates at $3m/s^2$ for 5s. Find final velocity and distance covered. 7+7

4. (a) State Newton's law of gravitation and write its mathematical form.

   (b) Calculate g at a depth of 500 m below Earth's surface (Radius of Earth = 6.37×10^6 m). 7+7

5. (a) Define modulus of rigidity and bulk modulus.

   (b) A rod of length 1m and area $2mm^2$ is stretched by 1mm under a load of 10kg. Calculate Young's modulus. 7+7

6. (a) Explain coefficient of thermal conductivity and its SI unit.

   (b) A rod conducts 200J of heat in 20s. Length = 50cm, area = $1cm^2$, temperature difference = 40°C. Find thermal conductivity. 7+7

7. Write short notes on *any four*: 3·5×4

   (a) Significant figures and types of errors

   (b) Law of conservation of linear momentum

   (c) Atmospheric pressure and gauge pressure

   (d) Modes of heat transmission

   (e) Stationary waves and their characteristics